Binary Outcomes

POLS 602
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Review

1. What does a variance inflation factor measure? How is it calculated?
2. Does multicollinearity bias the coefficients? What should you do about multicolinearity?
3. If one observation drives your result, what should you report?

Today

- What happens when the outcome is binary, and what does OLS estimate?
- How do we interpret that model, and where does it break down?
- How do we use the model to classify observations, and how do we evaluate the classifier?
- What are the alternatives, and when do we need them?

Binary Outcomes

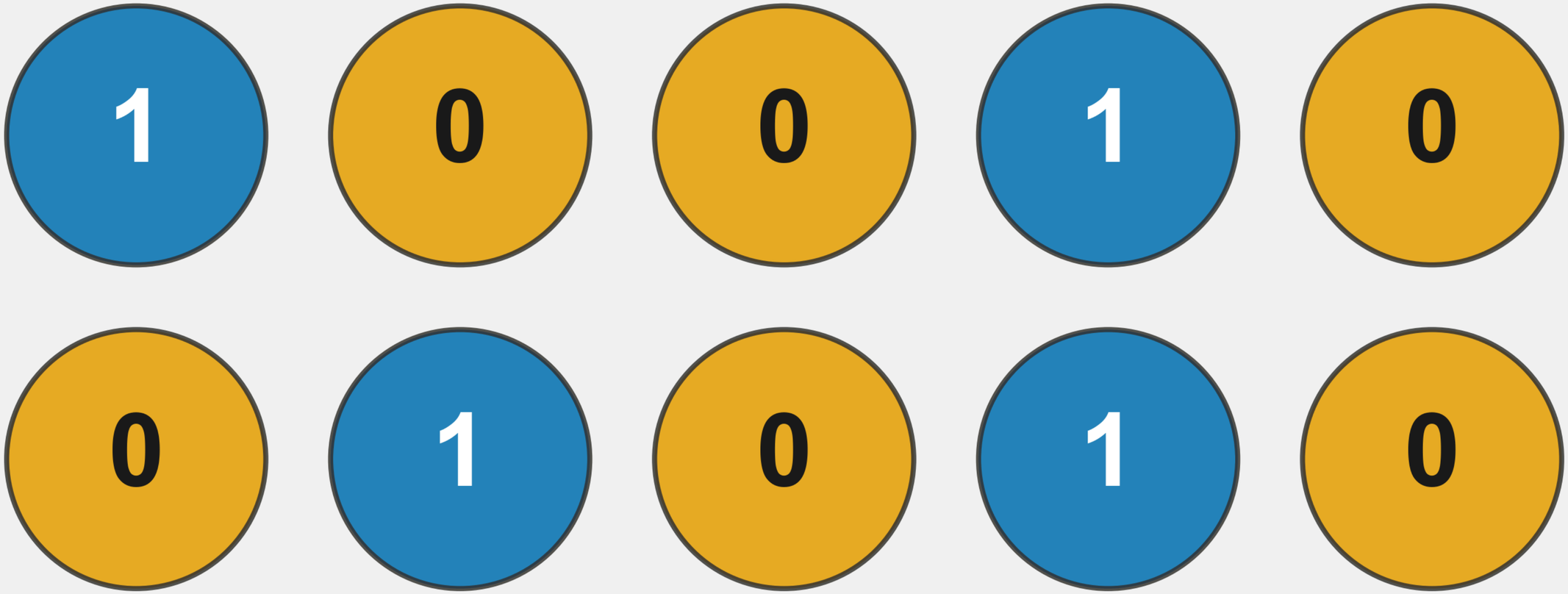
Binary Outcomes Are Everywhere

- **Binary outcome**: a variable that takes only two values, coded $Y_i = 1$ or $Y_i = 0$
- Did the person vote?
- Did the country experience civil war onset?
- Did the bill pass?
- Did the incumbent win?

Regression with Binary Outcomes

- A single outcome is 0 or 1, but regression estimates continuous values.
- What we can model is the probability that $Y_i = 1$ at each value of X_i
- We can do this because the probability of a binary event is a conditional mean

A Bag: Four 1s and Six 0s



Two Questions, One Answer

What is the average of the numbers on the marbles?

$$\frac{1+1+1+1+0+0+0+0+0+0}{10} = \frac{4}{10} = 0.4$$

If I draw one marble at random, what is the probability it is marked 1?

$$P(\text{draw a 1}) = \frac{4 \text{ marbles marked 1}}{10 \text{ marbles}} = 0.4$$

- A regression of a binary Y on X estimates a conditional mean, so it estimates a conditional probability

The Linear Probability Model

The Model

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

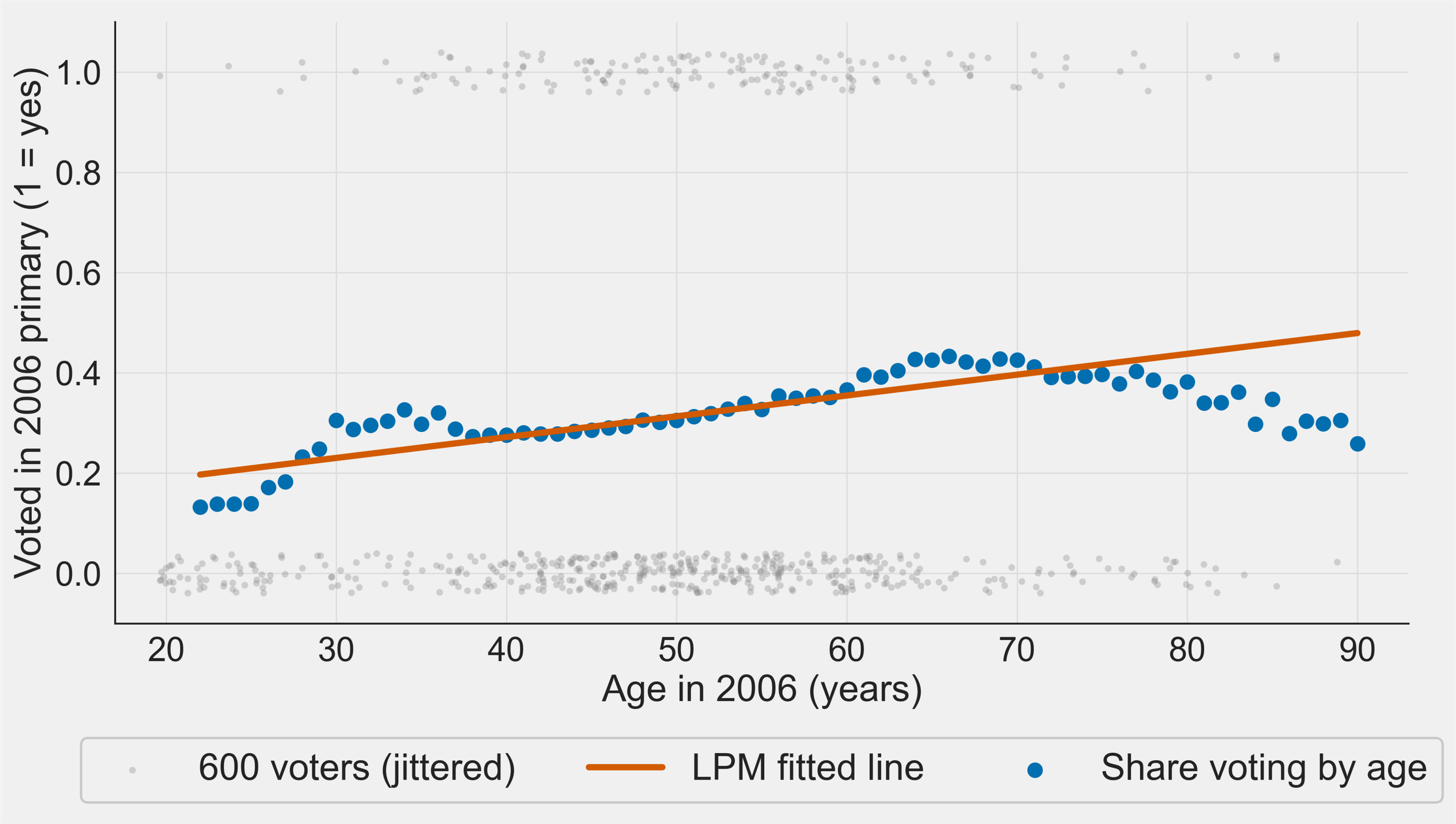
$$P(Y_i = 1 \mid X_i) = E[Y_i \mid X_i] = \beta_0 + \beta_1 X_i$$

- **Linear probability model (LPM)**: OLS with a binary dependent variable
- The model says the probability of $Y = 1$ is a linear function of the predictors
- With several predictors: $P(Y_i = 1 \mid \mathbf{x}_i) = \mathbf{x}_i^\top \boldsymbol{\beta}$

Why OLS Still Works

- OLS never required Y to be continuous
- Unbiasedness needs the conditional mean to be linear and $E[\epsilon \mid X] = 0$
- If $P(Y = 1 \mid X)$ really is linear in X, those conditions hold
- So $\hat{\beta}$ is unbiased for the effect of X on the probability

Example: Turnout and Age



Interpreting LPM Coefficients

$$\widehat{P}(\text{vote}_i = 1) = \widehat{\beta_0} + \widehat{\beta_1} X_i$$

$$\widehat{P}(\text{vote}_i = 1) = 0.106 + 0.0042 \times \text{age}_i$$

- $\widehat{\beta}_1$ is the change in the *probability* of $Y = 1$ for a one-unit increase in X
- Ten more years of age is associated with a $10 \times 0.0042 = 0.042$ increase, about 4 percentage points
- $\widehat{\beta}_0$ is the predicted probability when all predictors equal 0

Example: Multiple Regression LPM

	Estimate	Std. error
Civic Duty message	0.0186	0.0025
Hawthorne message	0.0257	0.0025
Neighbors message	0.0802	0.0025
Female	−0.0089	0.0016
Age (decades)	0.0388	0.0006
Household size	−0.0008	0.0011
Voted in 2004 primary	0.1468	0.0017
Intercept	0.0509	0.0046

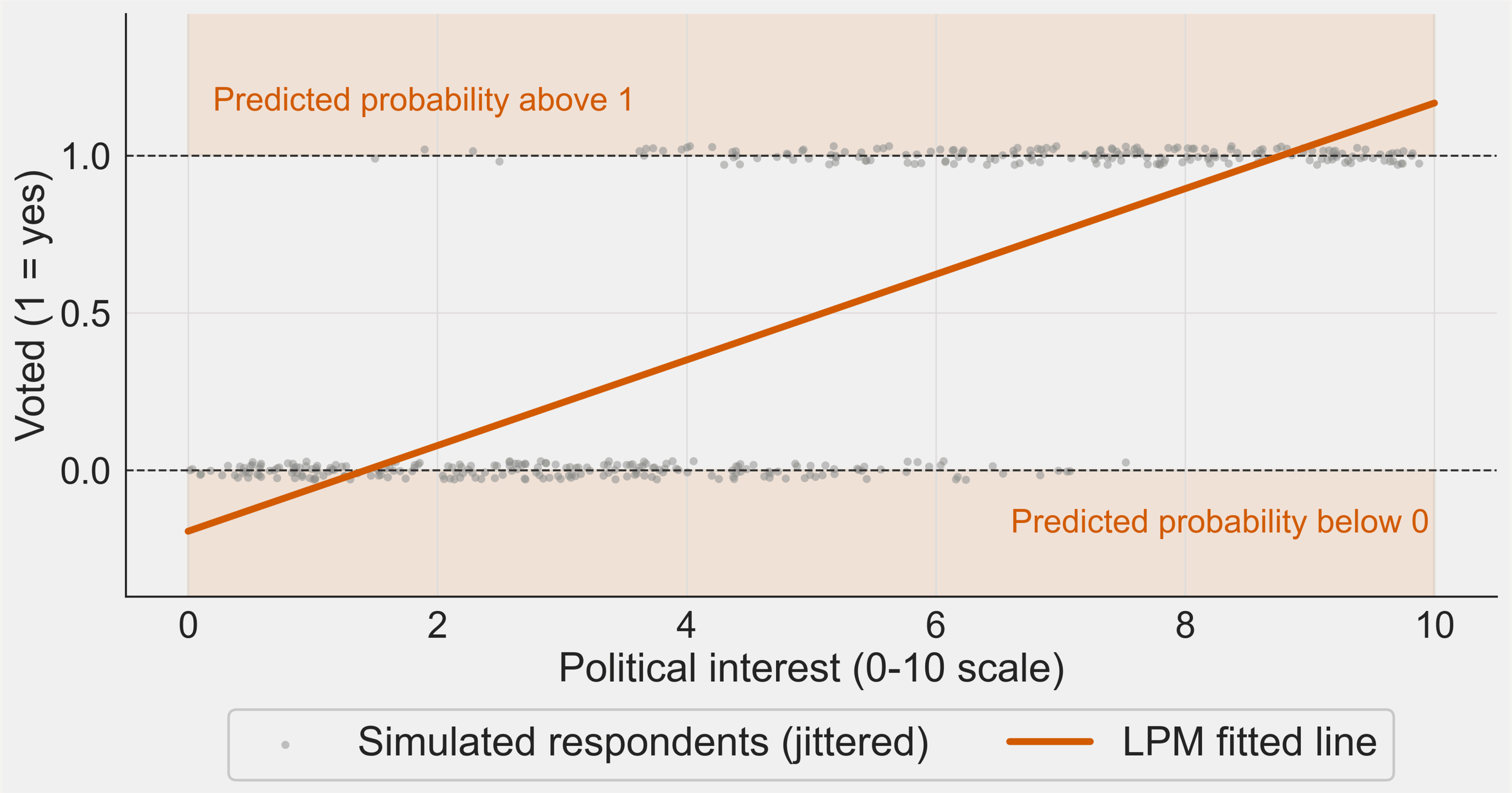
Outcome: voted in the 2006 primary (1 = yes). Omitted message category: Control. $n = 305{,}866$.

Effect as a Difference in Rates

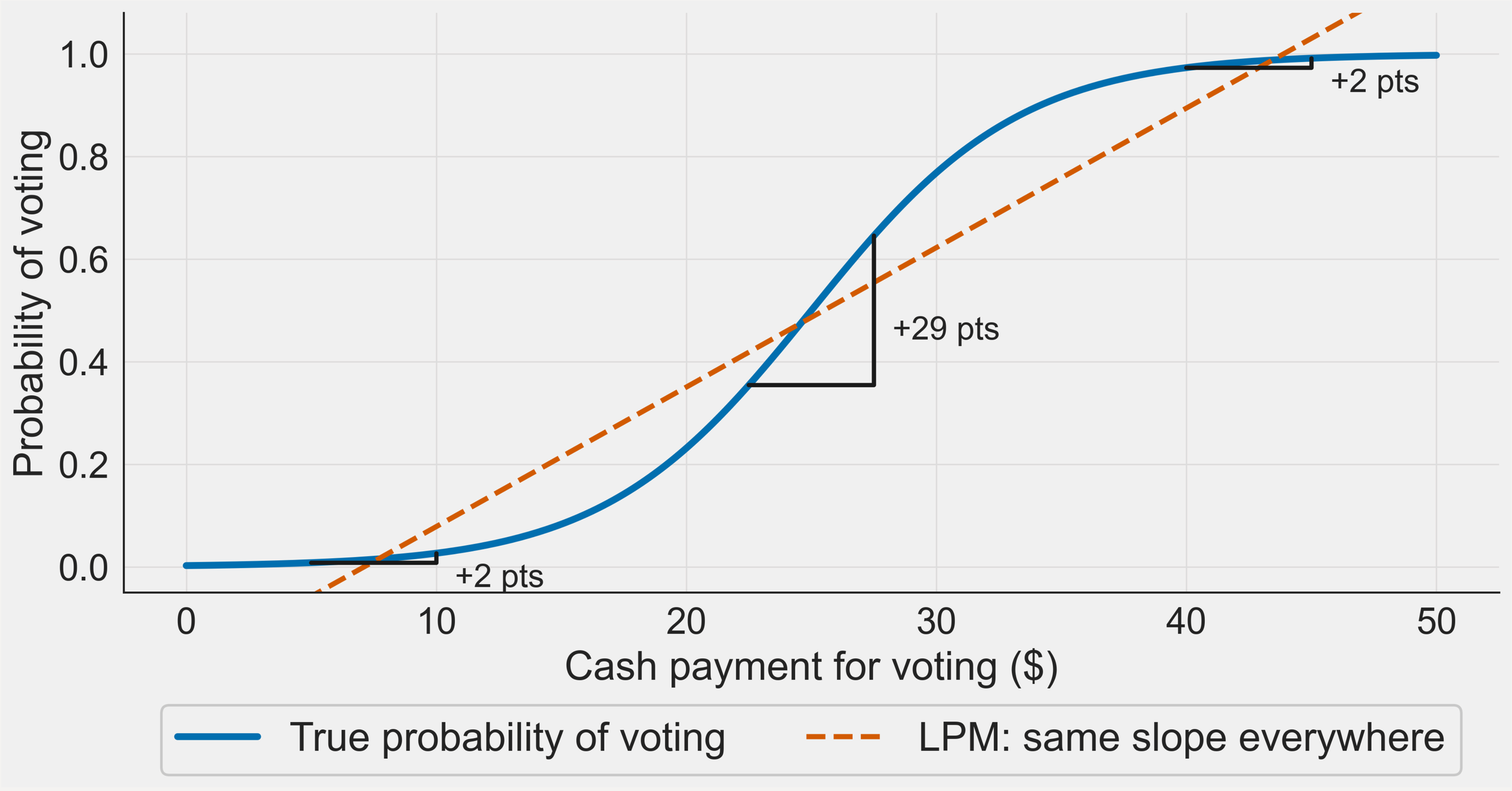
- With a single binary predictor, the LPM fits two group means exactly. Example, voter turnout experiment:
  - Control turnout: $\hat{\beta}_0 = 0.297$
  - Treatment turnout: $\hat{\beta}_0 + \hat{\beta}_1 = 0.378$
  - Effect: $\hat{\beta}_1 = 0.081$, a difference in proportions

Limits of the LPM

Problem 1: Impossible Predictions



Problem 2: Constant Effects



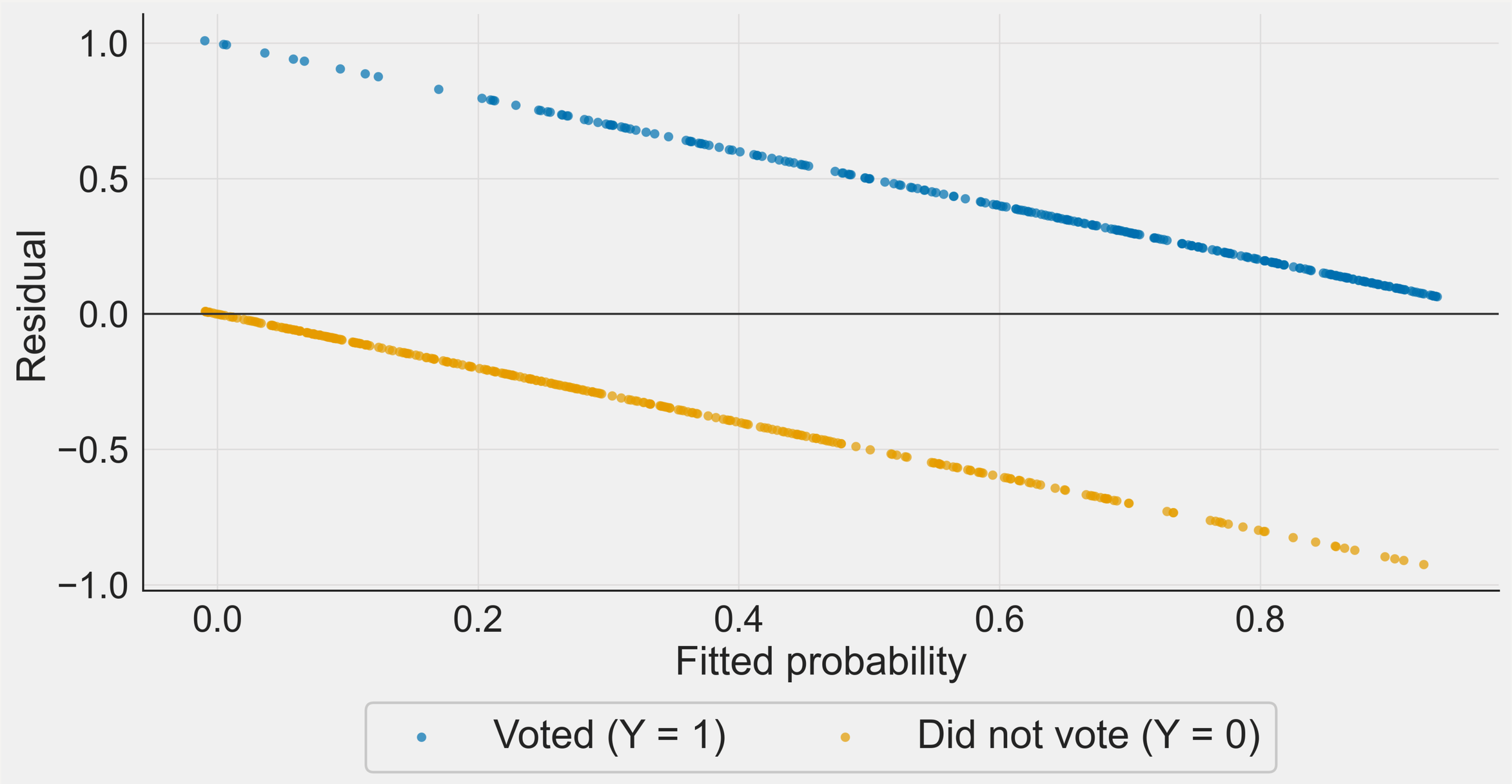
Problem 3: Two-Valued Errors

$$\epsilon_i = Y_i - p_i \quad \text{where } p_i = P(Y_i = 1 \mid X_i)$$

Outcome	Error	Probability
$Y_i = 1$	$1 - p_i$	p_i
$Y_i = 0$	$-p_i$	$1 - p_i$

- Each error can only be one of two numbers, so it cannot be normally distributed
- Which two numbers depends on p_i, so the error depends on X_i

Residuals Fall on Two Lines



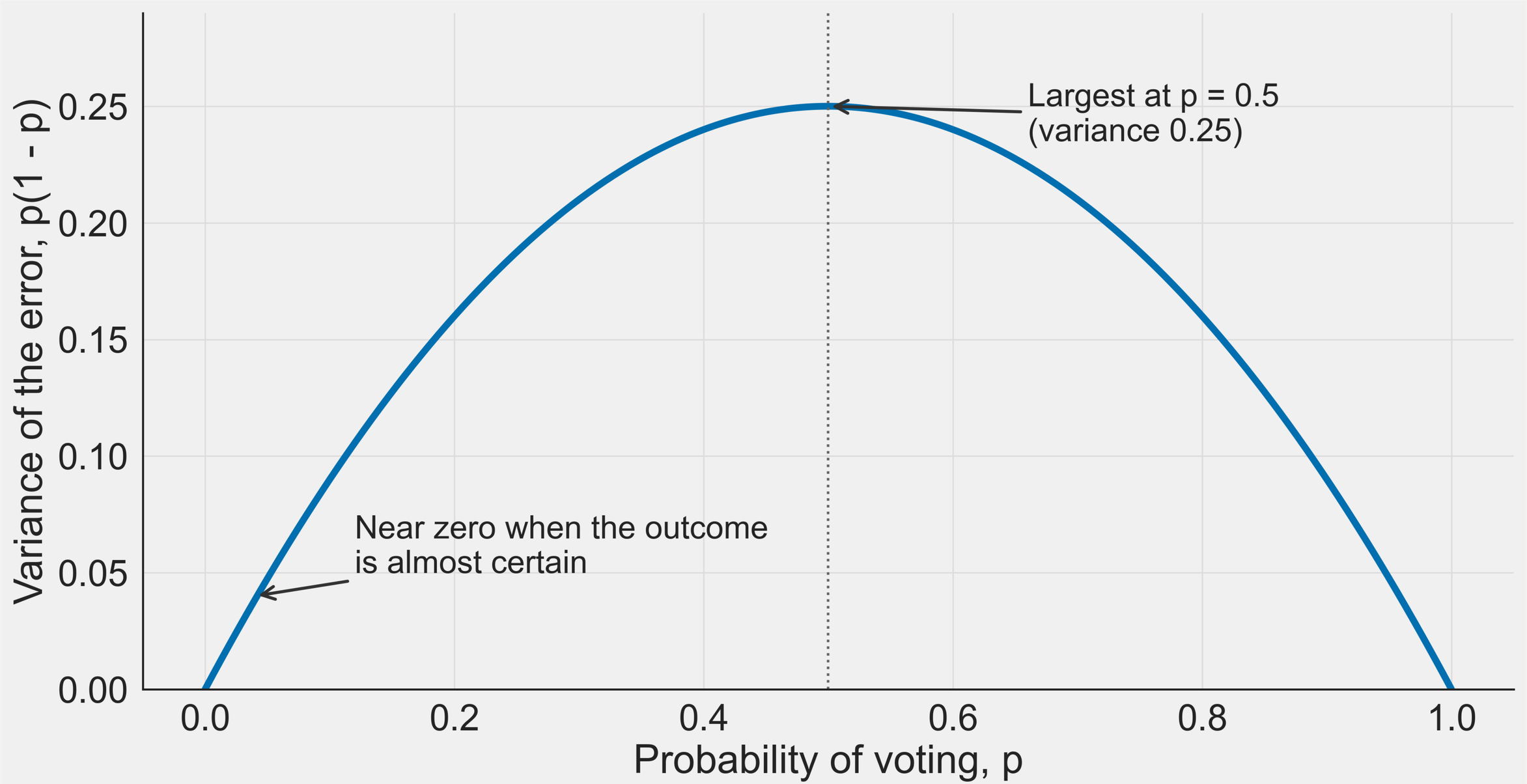
Problem 4: Heteroskedasticity

$$\text{Var}(\epsilon_i \mid X_i) = p_i(1 - p_i)$$

- Derivation: $\text{Var}(Y) = E[Y^2] - (E[Y])^2$
- Since Y is 0 or 1, $Y^2 = Y$, so $E[Y^2] = p$
- Therefore $\text{Var}(Y) = p - p^2 = p(1 - p)$

- The variance depends on p_i, and p_i depends on X_i
- So the variance of the errors changes with X_i, which violates the spherical errors assumption

Error Variance Peaks at p = 0.5



When Is the LPM Reasonable?

- Predictors with a binary or categorical effect, such as an experiment with a treatment dummy
- Fitted probabilities that stay well inside 0 and 1
- Interest is in the average effect, not the effect at a specific value
- Always use robust standard errors

Classification

Regression as a Classifier

- **Classification**: assigning each observation to one of two categories, such as liberal or conservative
- The goal is the right label for each person, not the size of a coefficient
- Useful when the label is known for some people and missing for others
- Plan: fit an LPM, turn each fitted probability into a label, then check the labels against the truth

Example: Liberal or Conservative?

- Simulated survey of 2,000 respondents
- Outcome: self-identified ideology, $Y_i = 1$ if liberal and 0 if conservative (35% liberal)
- Predictors: agreement, from 1 to 5, with statements on health coverage, gun laws, immigration, taxes on top earners, and abortion
- **Training set**: 1,000 respondents used to fit the model
- **Test set**: the other 1,000, held out to evaluate it

Step 1: Fit the LPM on the Training Set

$$P(\text{liberal}_i = 1 \mid \mathbf{x}_i) = \beta_0 + \beta_1 \text{Health}_i + \cdots + \beta_5 \text{Abortion}_i$$

	Estimate	Std. error
Health coverage	0.086	0.012
Gun laws	0.088	0.012
Immigration	0.071	0.012
Taxes on top earners	0.100	0.012
Abortion	0.037	0.012
Intercept	−0.841	0.032

Each coefficient is the change in the probability of being liberal for a one-point increase in agreement.

Step 2: Compute a Fitted Probability

A respondent answers Health 4, Guns 5, Immigration 3, Taxes 5, Abortion 4:

$$\begin{aligned}\hat{p} &= -0.841 + 0.086(4) + 0.088(5) + 0.071(3) \\ &\quad + 0.100(5) + 0.037(4) \\ &\approx 0.80\end{aligned}$$

- Each term is a coefficient times the respondent's answer
- The model estimates an 80% probability that this respondent is liberal
- A respondent who disagrees with all five statements (all answers equal to 1) gets $\hat{p} \approx -0.46$
- Fitted values from an LPM are not always valid probabilities

Step 3: Apply a Threshold

$$\widehat{Y}_i = \begin{cases} 1 \text{ (predict liberal)} & \text{if } \hat{p}_i \geq c \\ 0 \text{ (predict conservative)} & \text{if } \hat{p}_i < c \end{cases}$$

- The **threshold** c is the cutoff for a label of 1. The usual choice is $c = 0.5$
- With $c = 0.5$, we label someone liberal when the model says liberal is more likely than not

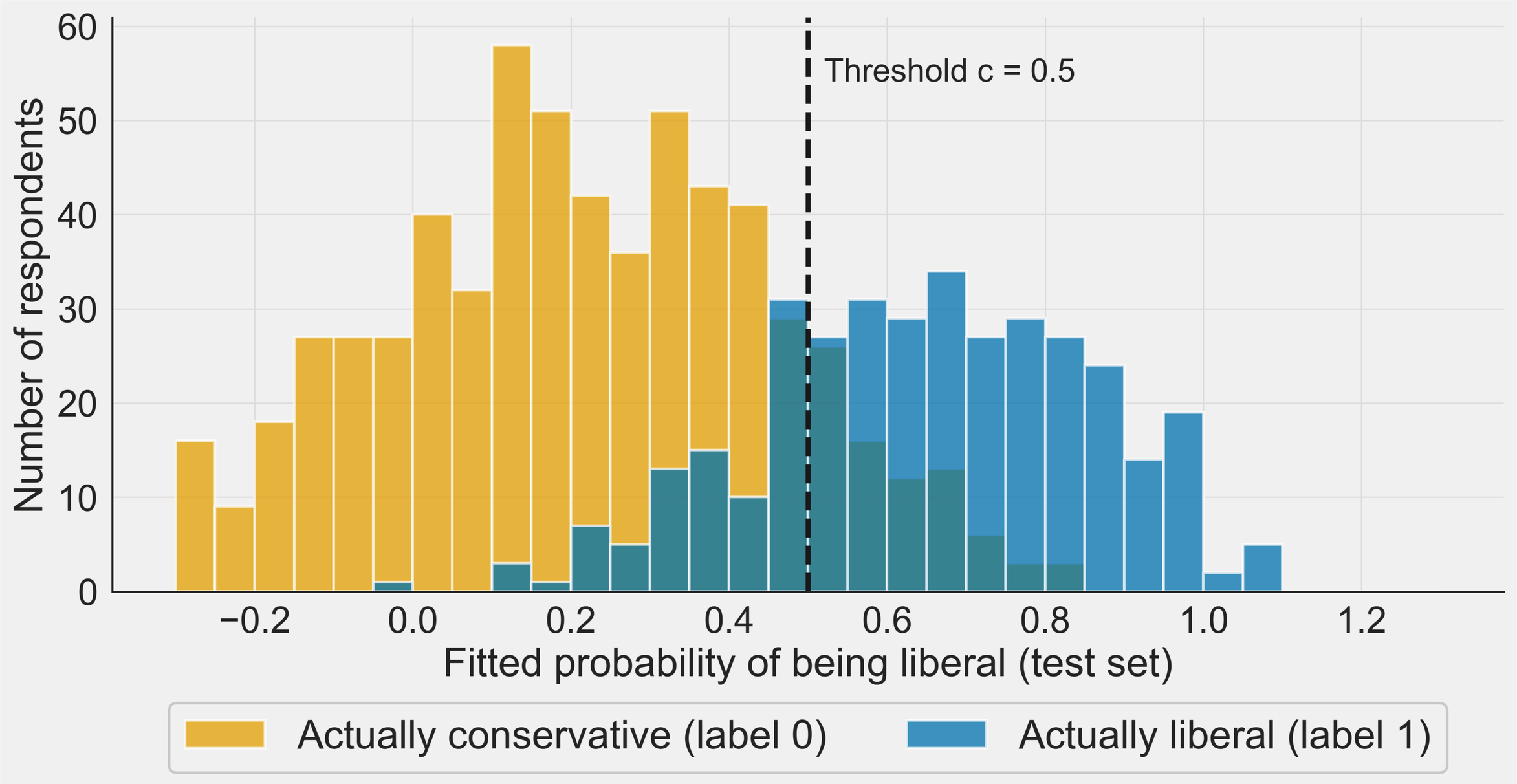
Four Respondents

Health	Guns	Immig.	Taxes	Abort.	$\hat{p}$	Pred.	Actual
4	5	3	5	4	0.80	Lib.	Lib.
4	2	3	1	3	0.10	Con.	Con.
4	4	5	3	2	0.58	Lib.	Con.
4	2	5	2	4	0.38	Con.	Lib.

The first five columns are agreement (1 to 5) with each statement. Pred. is the predicted label. Lib. is liberal and Con. is conservative.

- Rows 1 and 2 are correct. Rows 3 and 4 are mistakes
- A mistake can go in either direction

Liberals Mostly Score Above 0.5



Evaluating a Classifier

The Confusion Matrix

	Predicted 0	**Predicted 1**
Actual 0	True negative (TN)	False positive (FP)
Actual 1	False negative (FN)	True positive (TP)

- The **positive class** is the category coded 1. Here, liberal
- A **positive** is a respondent we label 1, and a **negative** is a respondent we label 0
- *True* or *false* says whether the label was correct

Our Confusion Matrix ($c = 0.5$)

	Predicted conservative	Predicted liberal	Total
Actual conservative	567 (TN)	79 (FP)	646
Actual liberal	86 (FN)	268 (TP)	354
Total	653	347	1,000

- 567 + 268 = 835 respondents are labeled correctly
- 79 conservatives are labeled liberal
- 86 liberals are labeled conservative

Accuracy, Sensitivity, Specificity

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{268 + 567}{1000} = 0.835$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} = \frac{268}{354} = 0.757$$

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{567}{646} = 0.878$$

- **Accuracy**: the share of all respondents labeled correctly
- **Sensitivity**: among actual liberals, the share we label liberal
- **Specificity**: among actual conservatives, the share we label conservative

Precision, Recall, and F1

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{268}{347} = 0.772$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{268}{354} = 0.757$$

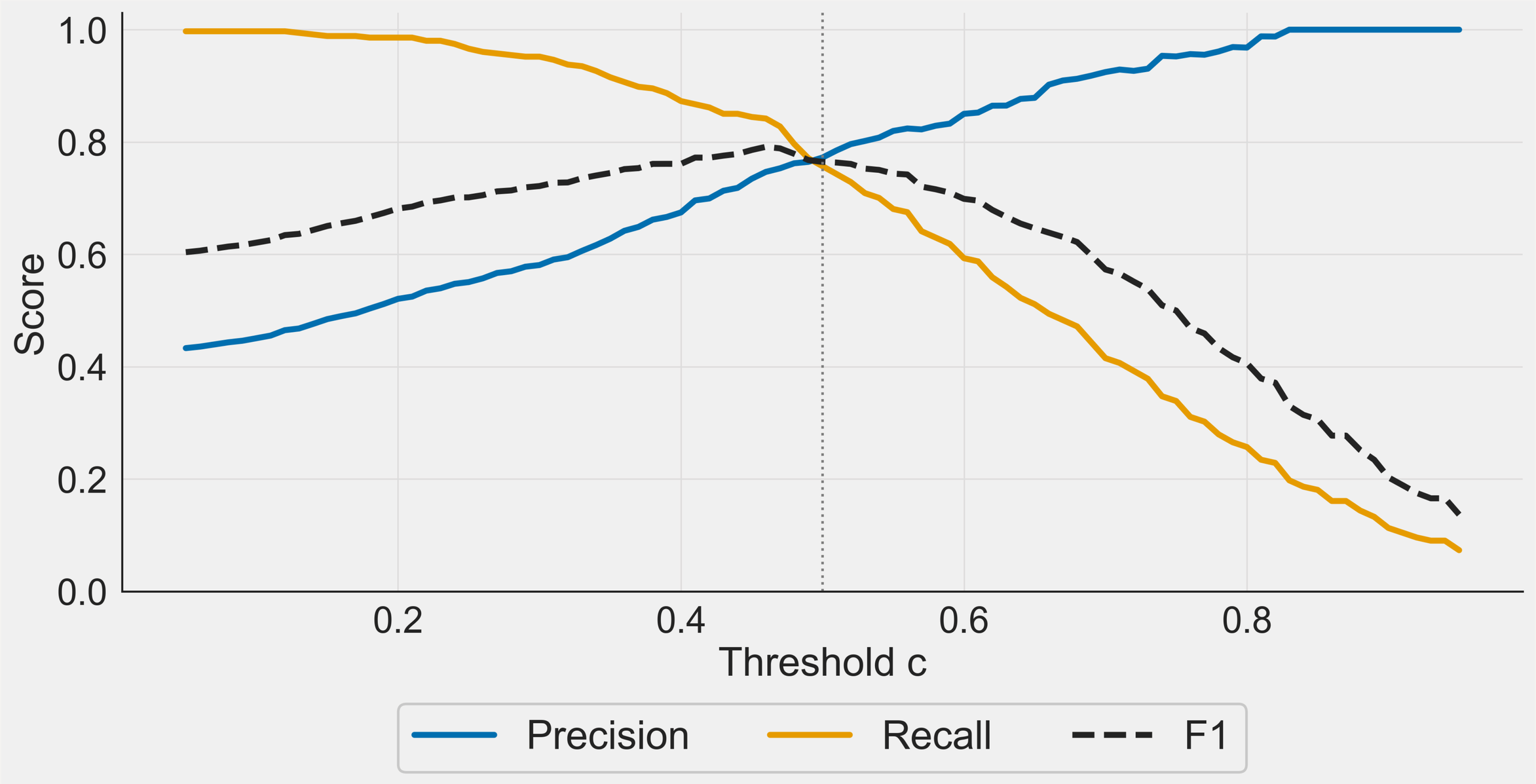
$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \times 0.772 \times 0.757}{0.772 + 0.757} = 0.765$$

Moving the Threshold

Threshold c	Accuracy	Precision	Recall	Specificity	F_1
0.3	0.740	0.581	0.952	0.624	0.722
0.5	0.835	0.772	0.757	0.878	0.765
0.7	0.781	0.925	0.415	0.981	0.573

- A lower threshold labels more respondents liberal: recall rises and precision falls
- A higher threshold labels fewer respondents liberal: precision rises and recall falls

Precision and Recall Trade Off



Cautions for Classification

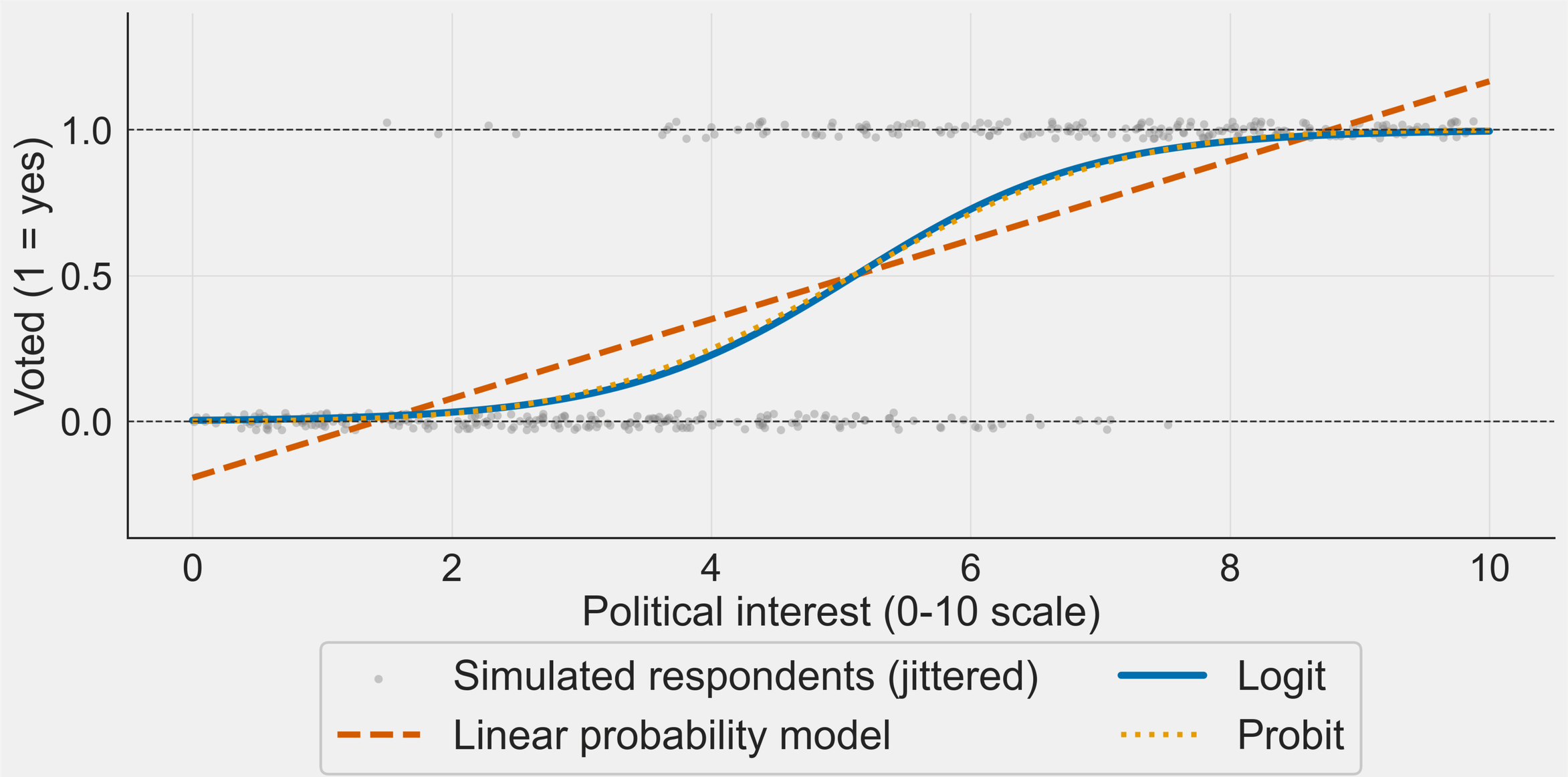
- Evaluate on a test set, not the data used to fit the model
- Compare accuracy to the baseline of always predicting the most common category
- Choose the threshold by the cost of each kind of mistake, not by default
- With an LPM, fitted values can leave 0 to 1, which does not affect labels but does limit them as probabilities

Logit and Probit

What We Want From a Binary Model

- Predicted probabilities that always stay between 0 and 1
- Effects that are small when the probability is near 0 or 1, and larger near 0.5
- Both properties come from an S-shaped curve

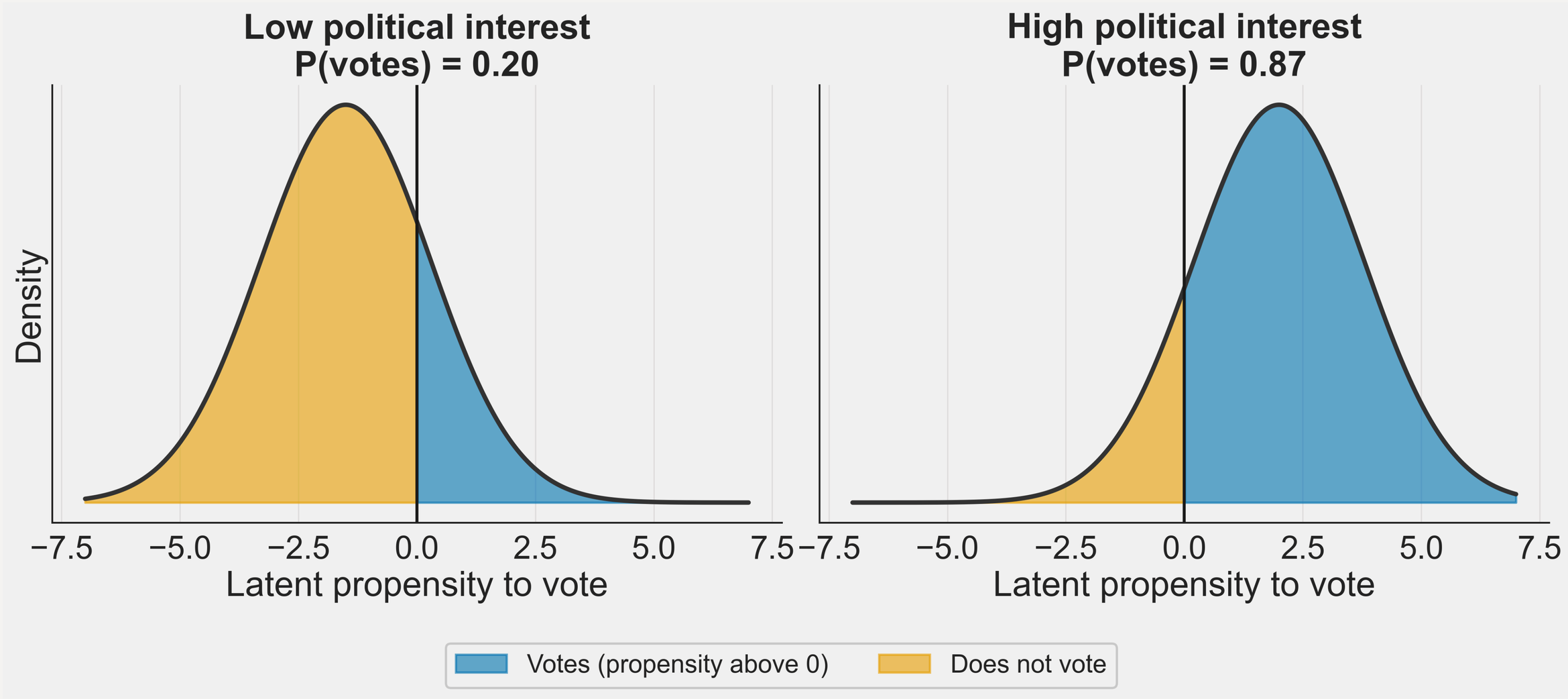
Logit and Probit Curves



The Latent Variable Idea

- Suppose each person has an unobserved *propensity* Y_i^* that depends on X_i plus noise
- We observe $Y_i = 1$ only if the propensity crosses a threshold, here 0
- The probability of $Y = 1$ is the area of the distribution above the threshold
- **Logit**: the noise has a logistic distribution
- **Probit**: the noise has a normal distribution

Area Above the Threshold



Logit and Probit Versus the LPM

- Estimated by maximum likelihood, not OLS
- Coefficients are not changes in probability. The effect on the probability depends on the value of X
- Predicted probabilities always stay between 0 and 1
- The sign and significance of a coefficient are directly interpretable

Putting It Together

Summary

- The mean of a binary variable is a probability, so OLS on a binary outcome estimates a conditional probability
- The LPM can predict outside 0 and 1 and assumes a constant effect, so use robust standard errors and check fitted values
- A fitted probability and a threshold turn the LPM into a classifier, which we judge with a confusion matrix, sensitivity, specificity, precision, and F1
- Logit and probit use an S-shaped curve from a latent variable model, which keeps probabilities in bounds

Key Definitions

Key Definitions

- **Binary outcome**: a variable with two values, coded 1 and 0
- **Linear probability model (LPM)**: OLS with a binary dependent variable, where the fitted value is the estimated probability that $Y = 1$
- **Saturated model**: a model with a free parameter for every group mean, such as a single dummy variable
- **Classification**: assigning each observation to a category, such as liberal or conservative
- **Classification rule**: predict $Y = 1$ when the fitted probability is at least a threshold c

Key Definitions

- **Training set and test set**: the data used to fit a model, and separate data held out to evaluate it
- **Confusion matrix**: a table of actual outcomes against predicted outcomes
- **Accuracy**: the share of all observations labeled correctly
- **Sensitivity (recall)**: the share of actual 1s that are predicted to be 1
- **Specificity**: the share of actual 0s that are predicted to be 0

Key Definitions

- **Precision**: the share of observations predicted to be 1 that are actually 1
- **F1 score**: the harmonic mean of precision and recall, high only if both are high
- **Logit and probit**: models for binary outcomes that pass a linear index through an S-shaped function so probabilities stay between 0 and 1
- **Latent variable**: an unobserved continuous variable that determines the observed outcome through a threshold